

step4_prediction

November 8, 2025

```
[2]: #Import libraries and load data/model

import pandas as pd
import joblib

from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, f1_score, classification_report
```

```
[3]: #Reload data and model artifacts
train = pd.read_csv("Load_model/train_temp.csv")
live = pd.read_csv("Load_model/live_temp.csv")
X = pd.read_csv("preprocess_data/X_temp.csv")
y = pd.read_csv("preprocess_data/y_temp.csv").squeeze()
grid_search = joblib.load("preprocess_data/grid_rf.pkl")

print("Data and model loaded.")
```

Data and model loaded.

```
[4]: #Get best model and evaluate on validation set

# Split again for validation evaluation
X_train, X_val, y_train, y_val = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

best_rf = grid_search.best_estimator_

y_val_pred = best_rf.predict(X_val)
```

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[5]: print("Validation Accuracy:", accuracy_score(y_val, y_val_pred))
print("Validation F1:", f1_score(y_val, y_val_pred))
print("\nClassification Report:\n", classification_report(y_val, y_val_pred))
```

Validation Accuracy: 0.8246411483253588

Validation F1: 0.7981272376755716

Classification Report:

	precision	recall	f1-score	support
0.0	0.87	0.82	0.84	2427
1.0	0.77	0.83	0.80	1753
accuracy			0.82	4180
macro avg	0.82	0.82	0.82	4180
weighted avg	0.83	0.82	0.83	4180

```
[6]: #Retrain best model on full data
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best_rf.fit(X, y)
print("Best model retrained on full training data.")
```

Best model retrained on full training data.

```
[7]: #Predict on live dataset and save output
```

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# Ensure 'id' column exists
if 'id' not in live.columns:
    live = live.reset_index().rename(columns={'index': 'id'})

# Match training features
live_features = live[X.columns]

# Predict
preds = best_rf.predict(live_features)

# Build output
output = pd.DataFrame({
    'id': live['id'],
    'prediction': preds
})

GROUP_NAME = "G19" # Change if needed
filename = f"{GROUP_NAME}_predictions.live.csv"

output.to_csv(f"test_data/{filename}", index=False)

print(f"Predictions saved as {filename}")
display(output.head())
```

Predictions saved as G19_predictions.live.csv

	id	prediction
0	6111	0.0
1	11214	0.0
2	5554	1.0
3	25131	1.0
4	14324	1.0